

Lab 11 Template

Part 01:

1. Take a screenshot of your decryption capturing the AAD data being displayed.
(10 points)

```
cpre3310@cpre3310:~/lab11/part01$ python3 part01_02_chacha20poly1305_skel.py decrypt part01_key.bin part01_nonce.bin part01_ciphertext.bin part01_plaintext_generated.txt part01_aad.bin
Authenticated AAD (ascii): netid=pulkit,lab=11,date=4/18/26,ip=52.126.35.1
File 'part01_ciphertext.bin' decrypted successfully to 'part01_plaintext_generated.txt'.
Decrypted content (ASCII): Have you heard of that new band "1023 Megabytes"? They're pretty good, but they don't have a gig just yet.
cpre3310@cpre3310:~/lab11/part01$
```

2. Upload your `part01_02_chacha20poly1305_skel.py` and your `part01_aad.bin` to Canvas.
(10 points, 5 points each item)

Part 02:

3. Take a screenshot of the decryption command and output including the AAD information.
(10 points)

```
cpre3310@cpre3310:~/lab11/part02$ python3 ../part01/part01_02_chacha20poly1305_skel.py decrypt part02_key.bin part02_nonce.bin part02_ciphertext.bin part02_plaintext_generated.txt part02_aad.bin
Authenticated AAD (ascii): msg-type=assignment;netid=jrursch;course=CprE3310;lab=11;part=2
File 'part02_ciphertext.bin' decrypted successfully to 'part02_plaintext_generated.txt'.
Decrypted content (ASCII): I tried to commit to something serious. Git said I had merge issues.
cpre3310@cpre3310:~/lab11/part02$ cat part02_aad.bin
msg-type=assignment;netid=jrursch;course=CprE3310;lab=11;part=2
cpre3310@cpre3310:~/lab11/part02$ ls -al part02_plaintext_generated.txt
-rw-rw-r-- 1 cpre3310 cpre3310 69 Apr 18 01:30 part02_plaintext_generated.txt
```

4. Take a screenshot of both your file sizes to include in your lab report.
(10 points, 5 points each item)

```
cpre3310@cpre3310:~/lab11/part02$ ls -al part02_plaintext_generated.txt
-rw-rw-r-- 1 cpre3310 cpre3310 69 Apr 18 01:30 part02_plaintext_generated.txt
cpre3310@cpre3310:~/lab11/part02$ ls -al part02_ciphertext.bin
-rw----- 1 cpre3310 cpre3310 85 Apr 17 23:42 part02_ciphertext.bin
cpre3310@cpre3310:~/lab11/part02$
```

5. Take a screenshot of your authentication tag (last 16 bytes) to include in your lab report.
(10 points)

```
cpre3310@cpre3310:~/lab11/part02$ tail -c 16 part02_ciphertext.bin | xxd
00000000: 6c21 707d 80f8 9a34 5449 4d58 5ee7 f56e  l!p}...4TIMX^...n
cpre3310@cpre3310:~/lab11/part02$
```

Part 03:

6. Take a screenshot of the command and output including the error message for the flipped bit and include it in your lab report.

(10 points)

```
cpre3310@cpre3310:~/lab11/part03$ python3 ../part01/part01_02_chacha20poly1305_skel.py decrypt part03_key.bin part03_non  
ce.bin part03_ciphertext.bin part03_plaintext_generated.txt part03_aad.bin  
DECRYPTION FAILED: authentication tag verification failed (tamper/wrong inputs).
```

Part 04:

7. Take a screenshot of the command and output including the error message for the changed AAD information and include it in your lab report.

(10 points)

```
cpre3310@cpre3310:~/lab11/part04$ python3 ../part01/part01_02_chacha20poly1305_skel.py decrypt part04_key.bin part04_non  
ce.bin part04_ciphertext.bin part04_plaintext_generated.txt part04_aad_wrong.bin  
DECRYPTION FAILED: authentication tag verification failed (tamper/wrong inputs).  
cpre3310@cpre3310:~/lab11/part04$
```

8. Explain the difference between confidentiality and integrity in your own words, using this lab as your example.

(10 points, 5 points each confidentiality, 5 points integrity)

Confidentiality basically means keeping a message secret from anyone who should not read it and in this lab, ChaCha20 provides confidentiality by encrypting the plaintext so that it becomes harder to understand the message without the correct key and nonce.

Over here, integrity basically means making sure that the message and data have not been changed basically and in this lab, Poly1305 basically provides integrity by generating an authentication tag over the ciphertext and AAD, so that's why when we changed one bit in the ciphertext in part03, or basically used the wrong AAD in part04, the decryption failed because the tag no longer matched.

9. What benefits does an integrated AEAD design provide beyond convenience?

(10 points)

An integrated AEAD design basically gives confidentiality, integrity and authenticity in 1 construction, which reduces the chances of implementation mistakes and basically the older designs like Encrypt-then-MAC required developers to combine separate encryption and authentication steps correctly, and mistakes in ordering or verification could basically create security problems and AEAD also authenticates associated data such as headers or metadata, so that any important unencrypted information cannot be modified without detection.

10. Compare AES-GCM to ChaCha20Poly1305. What are some similarities and differences?

(10 points)

AES-GCM and ChaCha20-Poly1305 are basically similar since both are AEAD ciphersuites that provide confidentiality, integrity, and authenticity. Moreover, both use a nonce, encrypt the plaintext, and produce an authentication tag that basically detects tampering.

Over here, the main difference is basically the underlying encryption method because AES-GCM uses AES as a block cipher with Galois/Counter Mode, while ChaCha20-Poly1305 basically uses the ChaCha20 stream cipher with the poly 1305 authenticator. Secondly, AES-GCM is often very fast on systems with hardware AES support, while ChaCha20-Poly1305 tends to perform especially well in software and on devices without any special AES hardware, and hence ChaCha20-Poly1305 is also often viewed as a stronger choice for consistent performance across different platforms.